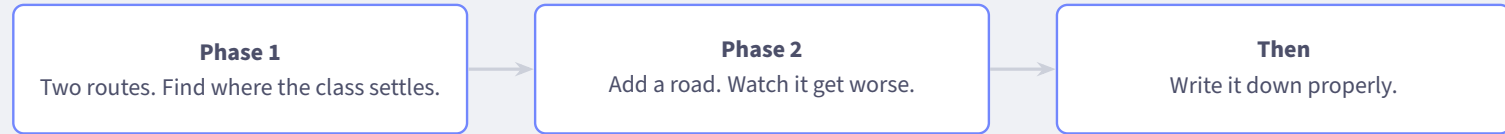


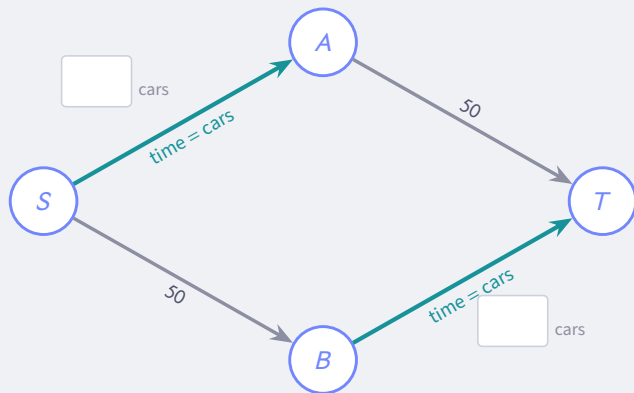
Routing games

Braess's paradox, in the room



Drivers in the room:

Date:



Top route

$$\boxed{} + 50 = \boxed{}$$

Bottom route

$$\boxed{} + 50 = \boxed{}$$

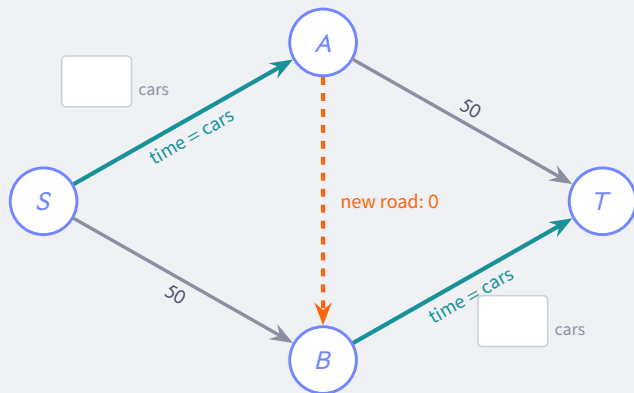
Stand on your route. Re-choose if the other one is quicker. Stop when nobody moves.

Round	On top	On bottom	Top time	Bottom time	Who moves?
1					
2					
3					
4					

If one driver switches from top to bottom, their time goes from to .

So this is a flow.

Is it also the best the class could do?



$S \rightarrow A \rightarrow T$

$S \rightarrow B \rightarrow T$

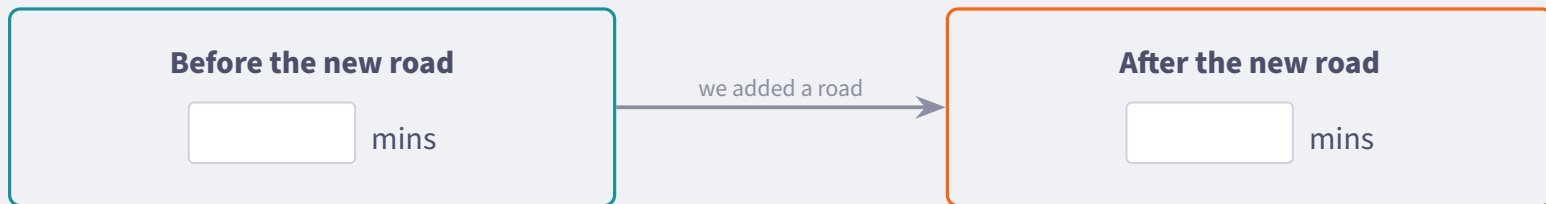
$S \rightarrow A \rightarrow B \rightarrow T$

Re-choose again. The new road is free, so who would not take it?

$S \rightarrow A$: cars = mins + $A \rightarrow B$: 0 mins + $B \rightarrow T$: cars = mins

Everyone's time: mins

Nobody wants to leave: going back to $S \rightarrow A \rightarrow T$ would cost + 50 = .



One more car joins a small road that already carries x cars.

It costs *me* minutes.

It costs *everyone else* minutes.

I pay the first and ignore the second. The road's total time is

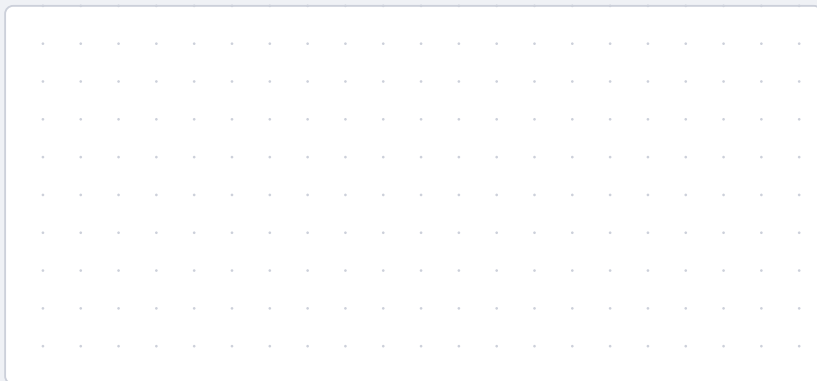
$x c(x) =$

and one more car really adds $\frac{d}{dx}(x c(x)) =$

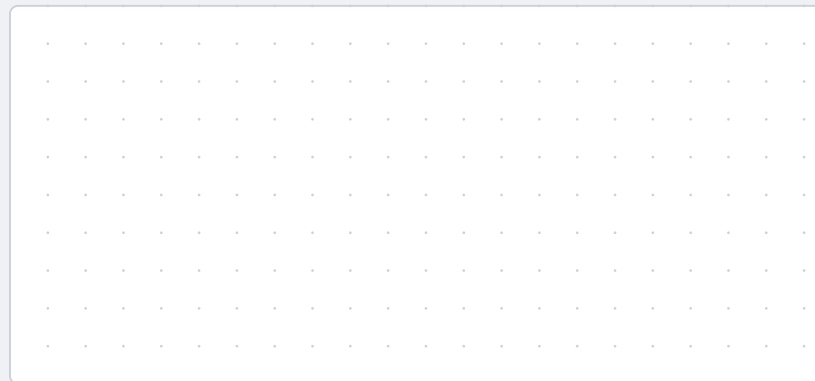
A routing game has a flow f on each route, and a cost

$$C(f) = \sum_e f_e c_e(f_e)$$

Our network, *before* the new road:



Our network, *after* the new road:



Left alone: the Nash flow $\tilde{f}$, and each driver's time:

$$\text{Price of anarchy} = \frac{C(\tilde{f})}{C(f^*)} = \frac{\boxed{}}{\boxed{}} = \boxed{}$$

What we just did in the room is **Question 1** on the Routing Games page, worth 25 marks.

- (a) What a Nash flow is here. [3]
- (b) The Nash flow and each driver's time. [5]
- (c) Adding $A \rightarrow B$: show all 40 on $S \rightarrow A \rightarrow B \rightarrow T$ is a Nash flow. [7]
- (d) Compare the times, and explain it through the congestion each driver ignores. [7]
- (e) Compute the price of anarchy. [3]

vknights.org/gt/topics/routing-games.html